

Job_Assistant_Overview

Project Overview: AI-Powered Job Application Assistant

Welcome to your first AI project!

This assignment challenges you to build an AI-Powered Job Application Assistant.

Vision: To build a tool that fundamentally streamlines the job application process, giving users back their time and providing them with an AI-powered co-pilot to land their dream job. Please see if you can make this browser extension or web-application rather than a desktop solution.

Core Features (The "What"):

1. **Autofill Agent:** Securely stores a user's profile information (name, contact info, work history, education, etc.) and intelligently fills out application forms across different websites.
2. **Resume-to-JD Scoring Agent:** Analyzes a user's uploaded resume against a specific job description (JD), provides a "match score," and generates actionable insights on how to tailor their resume to better fit the role.
3. **Tailored Answer Agent:** Analyzes the job description and the user's profile to generate high-quality, personalized answers to common application questions like, "Why are you a good fit for this role?" or "What is your biggest strength?".
4. **Dashboard :** The ability for a single screen to be able to walk the user(s) through where their application has been. Potential values: Not Submitted yet, Submitted, Received initial Response, Interview Requested, Rejected after Interview, Onsite/Video Interview Requested, etc.. (feel free to come up with this as you see fit)

Various Role Involvement: Each role (Data Scientist, Frontend Engineer, Backend Engineer) has distinct responsibilities. See the tabs on the left side for role specific assignments.

- **Frontend Engineers (The User's Champion):** You are in charge of the entire user experience. Your goal is to create an intuitive, trustworthy, and even delightful interface for our users to interact with the AI agents. You will bring the "magic" to life.
- **Backend Engineers (The Engine & Infrastructure):** You should try to build the robust and scalable systems that power the entire application. You'll create the APIs, manage data, and ensure the AI agents can be deployed, monitored, and maintained effectively. You are the architects of the system.
- **Data Scientists (The Intelligence):** You will design and build the core intelligence of the product—the AI agents themselves. You will work on scraping data, structuring prompts for Large Language Models (LLMs), and processing the results to deliver valuable insights to the user.

Key Concepts to Understand:

- **AI Agent:** Think of an agent not just as an AI model, but as a system that can perceive its environment, reason about a goal, and take actions to achieve it. In our case, an agent's goal might be "analyze this resume against this job description," and its actions would be to use tools like a web scraper and an LLM.
- **LLM (Large Language Model):** This is the "brain" of your agent (e.g., GPT-4, Gemini). It's a powerful reasoning and text-generation engine. We don't train the model itself; we learn how to communicate with it effectively.
- **Prompt Engineering:** This is the skill of crafting precise instructions (prompts) to get the desired output from an LLM. A well-designed prompt is the difference between a useless response and a highly effective one. This is a responsibility for everyone, not just the data scientists.
- **Tool Use:** This is a core agentic design pattern. An agent doesn't do everything by itself. It uses "tools"—which are just functions you write—to interact with the outside world. For this project, your key tools will be:
 - A **web scraper** to get data from a job posting URL.
 - A **database function** to retrieve a user's profile.
- **Structured Output (JSON):** For an application to work reliably, the LLM can't just return random text. We must force it to respond in a predictable format like JSON. This is done through prompt engineering and is the critical "contract" between the data science logic and the backend server.
- **API (Application Programming Interface):** This is the set of rules and endpoints that the backend provides so the frontend can communicate with it. It's the connective tissue of the entire project. For example, the frontend will make an API call to `POST /api/resume/analyze` to trigger the analysis agent.
- **Agentic UX (User Experience for AI):** Designing for AI is different from traditional UX. Key principles include:
 - **Showing the work:** Let the user know the agent is working on their behalf.
 - **Allowing intervention:** Users must be able to edit or reject the AI's suggestions. The AI is a co-pilot, not the pilot.
 - **Collecting feedback:** Incorporating simple feedback mechanisms is essential for knowing if the agent is actually helpful.



Note: To assist with the development process, you are free to use any AI coding, feel free to use Azure AI Foundry and the free DeepSeek model.

- A. YT Deploy DeepSeek R1 using Azure AI Foundry and Build a Web Chatbot | No Charges for API Use = <https://www.youtube.com/watch?v=pj2knBX4S1w>
- B. MSFT = <https://azure.microsoft.com/en-us/pricing/purchase-options/azure-account> [Available only to new Azure customers = \$200 credit]
- C. MSFT Billing for Free Trial Account = <https://learn.microsoft.com/en-us/answers/questions/2283380/billing-for-free-trial-account>
- D. MSFT - DeepSeek R1 is now available on Azure AI Foundry and GitHub = <https://azure.microsoft.com/en-us/blog/deepseek-r1-is-now-available-on-azure-ai-foundry-and-github/>

Go through all of the links above so you have a good idea

Good luck, and remember to have fun learning!

Helpful Resources:

Here are some helpful resources that could be used in your journey.

- Youtube: [Google Launches an Agent SDK - Agent Development Kit](#)
- Youtube: [Google Agent Development Kit](#)
 - Code: <https://mer.vin/2025/05/google-adk-agents/>
- Youtube: [Agent Development Kit \(ADK\) Masterclass: Build AI Agents & Automate Workflows \(Beginner to Pro\)](#)
- Youtube: [Build an AI Agent From Scratch in Python - Tutorial for Beginners](#)
- Youtube: [SmolAgents: A Smol Library to Build Agents](#)
- Youtube: [Build Multi-Agent Systems with SmolAgents](#)

Office Hours *(for weeks 1-3):*

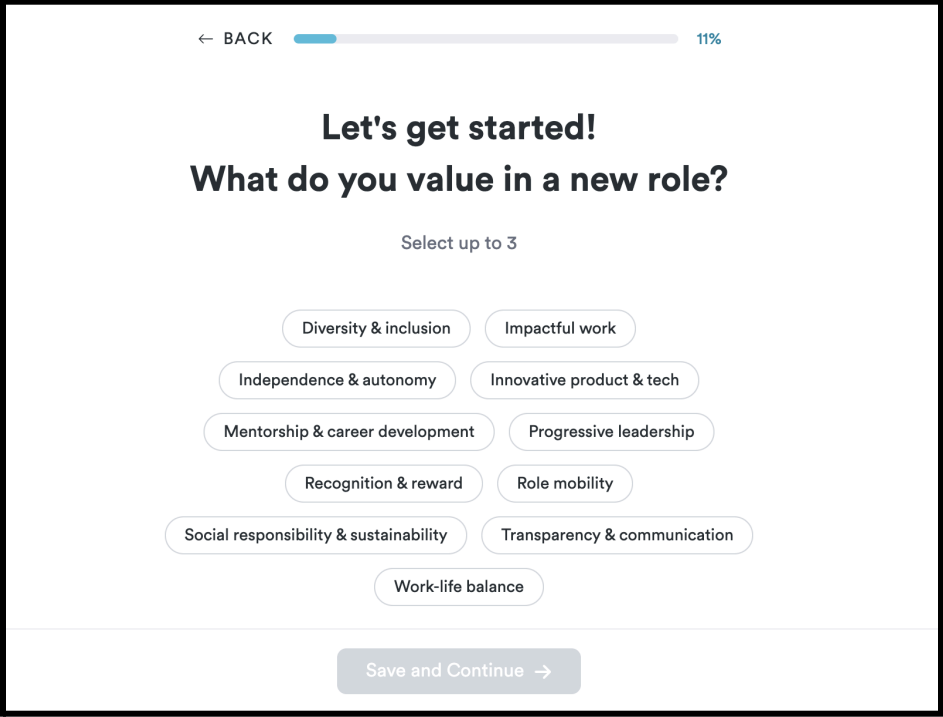
- Please look for announcements in Discord for exact dates/times of Office Hours.
- Use Office Hours to help with roadblocks rather than as a gatekeeper holding you back. This way you start coding and make progress while not waiting for scheduled weekly Office hour sessions.

Point of Reference:

[Simplify](#) is a good example of such UI/UX and features for your reference.

- It has [Chrome](#) and [Firefox](#) extensions.
- Feel free to go to that website for UI/UX inspiration.

Screens during onboarding asking about Preferences:



← BACK 11%

Let's get started!

What do you value in a new role?

Select up to 3

Diversity & inclusion Impactful work

Independence & autonomy Innovative product & tech

Mentorship & career development Progressive leadership

Recognition & reward Role mobility

Social responsibility & sustainability Transparency & communication

Work-life balance

Save and Continue →

What kinds of roles are you interested in?

Select up to 5



Search by job title

Technical & Engineering

- Aerospace Engineering
- AI & Machine Learning
- Architecture & Civil Engineering
- Data & Analytics
- Developer Relations
- DevOps & Infrastructure
- Electrical Engineering
- Engineering Management
- Hardware Engineering
- IT & Security
- Mechanical Engineering
- Process Engineering
- QA & Testing
- Quantitative Finance
- Quantum Computing
- Sales & Solution Engineering
- Software Engineering

Finance & Operations & Strategy

- Accounting
- Business & Strategy
- Consulting
- Finance & Banking
- Growth & Marketing
- Operations & Logistics
- Product
- Real Estate
- Retail
- Sales & Account Management

Creative & Design

- Art, Graphics & Animation
- Audio & Sound Design
- Content & Writing
- Creative Production
- Journalism
- Social Media
- UI/UX & Design

Education & Training

- Education
- Training

Legal & Support & Administration

- Administrative & Executive Assistance
- Clerical & Data Entry
- Customer Experience & Support
- Legal & Compliance
- People & HR
- Security & Protective Services

Life Sciences

- Biology & Biotech
- Lab & Research
- Medical, Clinical & Veterinary



Michael, Founder

Choose specializations to refine your preferences! We'll show you job matches to your specializations first, and then your general role types.

AI & Machine Learning (Select the most relevant specializations for you)

- ☐ AI Research
- ☐ Applied Machine Learning
- ☐ Computer Vision
- ☐ Conversational AI & Chatbots
- ☐ Deep Learning
- ☐ Natural Language Processing (NLP)
- ☐ Robotics & Autonomous Systems
- ☐ Speech Recognition

Save and Continue →

Where would you like to work?

United States

☐ Select all in United States

☐ Atlanta	☐ Austin
☐ Baltimore	☐ Boston
☐ Charlotte	☐ Chicago
☐ Dallas	☐ Denver
☐ Las Vegas	☐ Los Angeles
☐ Miami	☐ New York City
☐ Philadelphia	☐ Phoenix
☐ Portland	☐ Remote in USA Remote
☐ San Diego	☐ San Francisco Bay Area
☐ Seattle	☐ Washington D.C.

[+ Add Location](#)

Canada

☐ Select all in Canada

☐ Montreal	☐ Ottawa
☐ Quebec City	☐ Remote in Canada Remote
☐ Toronto	☐ Vancouver
☐ Winnipeg	

[+ Add Location](#)

United Kingdom

☐ Select all in United Kingdom

☐ Birmingham	☐ Liverpool
☐ London	☐ Manchester
☐ Remote in UK Remote	

[+ Add Location](#)

Australia

☐ Select all in Australia

Save and Continue →

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What level of roles are you looking for?

Select up to 2

- ☐ Internship
- ☐ Entry Level & New Grad
- ☐ Junior (1 to 2 years)
- ☐ Mid-level (3 to 4 years)
- ☐ Senior (5 to 8 years)
- ☐ Expert & Leadership (9+ years)

Save and Continue →

← BACK 40%

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- ☒ Senior (5 to 8 years)
- ☒ Expert & Leadership (9+ years)

Are you looking for a specific leadership role?

- ☐ Individual Contributor
- ☐ Manager
- ☒ I don't have a preference

Save and Continue →

← BACK 50%

What is your ideal company size?

☒ Unselect all sizes

☒ 1-10 employees

☒ 11-50 employees

☒ 51-200 employees

☒ 201-500 employees

☒ 501-1,000 employees

☒ 1,001-5,000 employees

☒ 5,001-10,000 employees

☒ 10,001+ employees

Save and Continue →

← BACK

60%

What industries are exciting to you?

✓ First, what industries are exciting to you?

Aerospace AI & Machine Learning

Automotive & Transportation Biotechnology Consulting

Consumer Goods Consumer Software Crypto & Web3

Cybersecurity Data & Analytics Defense Design

Education Energy Enterprise Software Entertainment

Financial Services Fintech Food & Agriculture Gaming

Government & Public Sector Hardware Healthcare

Industrial & Manufacturing Legal Quantitative Finance

Real Estate Robotics & Automation Social Impact

Venture Capital VR & AR

✗ Second, are there any industries you don't want to work in?

Aerospace AI & Machine Learning

Automotive & Transportation Biotechnology Consulting

Consumer Goods Consumer Software Crypto & Web3

Cybersecurity Data & Analytics Defense Design

Education Energy Enterprise Software

Entertainment Financial Services Fintech

Food & Agriculture Gaming Government & Public Sector

Hardware Healthcare Industrial & Manufacturing

Legal Quantitative Finance Real Estate

Robotics & Automation Social Impact Venture Capital

VR & AR

Save and Continue →

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70%

What skills do you have or enjoy working with?

Select all that applies

♥ Heart a skill indicate that you'd prefer roles that utilize that skill!

Search all skills...

Business Analytics

Excel/Numbers/Sheets

Git

HTML/CSS

Java

MailChimp

MATLAB

Operations Research

Python

SEO

Zendesk

Selected skills

♥ Adobe Illustrator

X

🚫 Are there any skills you don't want to work with?

Skills to filter out

Save and Continue →

← BACK

80%

What is your minimum expected salary?



Michael, Founder

We'll only use this to match you with jobs and will not share this data

At least

\$0k

USD

Save and Continue →

[< BACK](#) 100%

Lastly, what's the status of your job search?

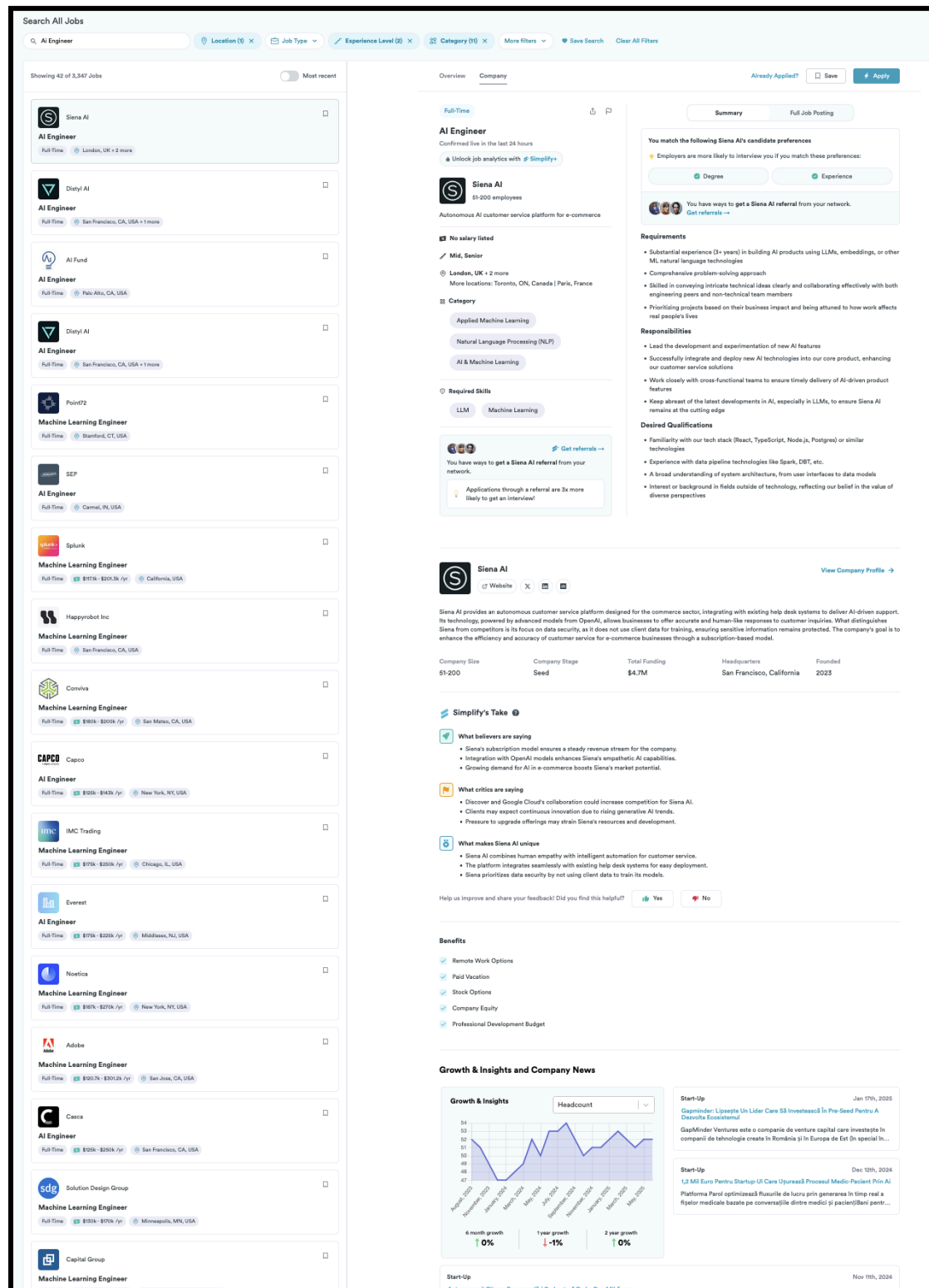
☐ Actively looking

☐ Not looking but open to offers

☐ Not looking and closed to offers

Save →

Results of Job Search example below:



Overview

Company

Already Applied?

Save

Apply

Full-Time

Share

Flag

AI Engineer

Confirmed live in the last 24 hours

Unlock job analytics with Simplify+



Siena AI

51-200 employees

Autonomous AI customer service platform for e-commerce

No salary listed

Mid, Senior

London, UK + 2 more

More locations: Toronto, ON, Canada | Paris, France

Category

Applied Machine Learning

Natural Language Processing (NLP)

AI & Machine Learning

Required Skills

LLM

Machine Learning



Get referrals →

You have ways to get a Siena AI referral from your network.

Applications through a referral are 3x more likely to get an interview!

Summary

Full Job Posting

You match the following Siena AI's candidate preferences

Employers are more likely to interview you if you match these preferences:

Degree

Experience



You have ways to get a Siena AI referral from your network. Get referrals →

Requirements

Substantial experience (3+ years) in building AI products using LLMs, embeddings, or other ML natural language technologies

Comprehensive problem-solving approach

Skilled in conveying intricate technical ideas clearly and collaborating effectively with both engineering peers and non-technical team members

Prioritizing projects based on their business impact and being attuned to how work affects real people's lives

Responsibilities

Lead the development and experimentation of new AI features

Successfully integrate and deploy new AI technologies into our core product, enhancing our customer service solutions

Work closely with cross-functional teams to ensure timely delivery of AI-driven product features

Keep abreast of the latest developments in AI, especially in LLMs, to ensure Siena AI remains at the cutting edge

Desired Qualifications

Familiarity with our tech stack (React, TypeScript, Node.js, Postgres) or similar technologies

Experience with data pipeline technologies like Spark, DBT, etc.

A broad understanding of system architecture, from user interfaces to data models

Interest or background in fields outside of technology, reflecting our belief in the value of diverse perspectives

Frontend Assignment

Frontend Engineer Assignment

As a Frontend Engineer, your goal is to be the user's champion. You will design and build the user experience for the AI Job Assistant, creating an intuitive, trustworthy, and effective interface for interacting with the AI agents. You will bring the "magic" of the AI to life for the user.

Phase 1: Pre-Office Hours Preparation (1 Week)

The goal of this phase is to move from idea to a design plan. You need to understand the unique challenges of designing for AI and translate that understanding into a visual and interactive blueprint for the application.

Actions:

- 1. Research Agentic UX:** Review these resources and synthesize their key principles. How should an application behave when an AI is performing tasks on the user's behalf?
 - Required Reading: [Microsoft's UX Design for Agents](#)
 - Required Reading: [Secrets of Agentic UX](#)
- 2. Map the User Journey:** Create user flow diagrams to visualize how a user will interact with each core feature. Think step-by-step:
 - Onboarding: How does a user securely create a profile and input their work history, skills, etc.?
 - Activation: How does the user trigger the Autofill, Resume Score, or Tailored Answer agents? Is it a browser sidebar? A pop-up?
 - Interaction & Feedback: How is AI-generated content (like resume suggestions or tailored answers) presented? How does the user approve, edit, or reject it? Crucially, how will you design a system for users to give feedback on the AI's quality (e.g., thumbs up/down)? This feedback loop is essential for improving the product.
- 3. Create Low-Fidelity Wireframes:** Using a tool like Figma, Balsamiq, or even just pen and paper, sketch out the primary UI components.
 - Profile/Settings Screen: Where the user's data lives.
 - Main Assistant Interface: The primary view when the assistant is active on a job page.
 - Resume Score Modal/View: The dashboard showing the match score and improvement tips.

- Generated Answer Component: The UI for showing suggested text and allowing the user to copy or edit it.

Office Hours (optional):

Use Office Hours to help with roadblocks rather than as a gatekeeper holding you back. This way you can start developing your code to make progress and no longer wait for Office hours.

Some things to think about:

- Think through your user flow and potentially create a small wireframe to help you think through screens.
- What could be some challenges you anticipate in implementing this design.

Phase 2: Development

Can start development even if not attend Office hours. Start focusing on aligning on the UX, your focus shifts to implementing the Discord bot interface using ReactJS/NextJS.

Actions:

Set Up the Project Environment: Initialize a new React.js or Next.js application. Set up your folder structure for components, services, and styles.

Build Static Components: Translate your wireframes into static React components. Focus on a clean, modern look and feel. Do not worry about connecting to APIs yet; you can use mock/dummy data to populate the UI.

- Build the User Profile Form.
- Build the Resume Analyzer component.
- Build the Generated Answer component.

Implement State Management: Use React Hooks (useState, useEffect, useContext) to manage the application's state.

Create API Service Functions: Create a dedicated file (e.g., services/api.js) to house your fetch calls. This keeps your components clean and separates the logic for communicating with the backend

Note: There are a few helpful resource videos to reference in the main [Job Assistant Overview](#) page.

Backend Assignment

Backend Engineer Assignment

As a Backend Engineer, you will build the engine of the AI Job Assistant. You will design and construct a secure, scalable, and observable system that powers the entire application. You are the architect of the infrastructure that enables the frontend and data science components to work together seamlessly.

Phase 1: Preparation

The goal of this phase is to design a robust backend architecture. You will define the technologies, data flows, and infrastructure needed to support the application's features now and in the future.

Actions:

1. **Research Agentic System Design:** To build a system that supports AI agents, you need to understand their unique requirements.
 - **Required Reading:** [The Agentic AI Handbook](#)
 - **Required Reading:** [4 AI Agent Strategies](#)
2. **Design the System Architecture:** Create a high-level architecture diagram using a tool like Lucidchart, draw.io, or even just boxes and arrows. Your diagram should show:
 - **The Client:** The Frontend application.
 - **API Layer:** Your Node.js/Express or Python/Flask application. Will you use an API Gateway?
 - **Core Logic:** Where will the "agent" logic (developed by the Data Science team) plug in? Will it be a separate service or integrated directly?
 - **Database:** Choose a database (e.g., PostgreSQL for structured data, MongoDB for flexibility).
 - **External Services:** Show the connection to a third-party LLM API (e.g., OpenAI, Gemini).
3. **Plan for Production Operations (*optional*):** Think beyond just writing code. How will this run in the real world?
 - **Deployment Strategy:** Focus initially to have this running locally.
 - Future think through: How will you package the application (e.g., Docker container)? Where do you plan to deploy it (e.g., AWS, Vercel, Heroku)?
 - **Logging & Observability:** What tools will you use for logging API requests, errors, and agent activity? How will you monitor system health (e.g., a /status health check endpoint)?

Office Hours (optional):

Use Office Hours to help with roadblocks rather than as a gatekeeper holding you back. This way you can start developing your code to make progress and no longer wait for Office hours.

Some things to think about:

- Your system architecture diagram.
- What is your high-level plan for deployment and logging?

Phase 2: Development

Can start development even if not attend Office hours. Start focusing on building the foundation. Your focus is on creating the API endpoints, setting up the database.

Actions:

1. **Set Up Project & Database:** Write a script to connect to your chosen database.
2. **Define Database Schemas:** Based on the Data Science team's data models, create the schemas or models for your database
3. **Build API Endpoints:** Create the core API endpoints. Initially, these can return mock data, but they should respect the data contracts you've agreed upon with the frontend team.
 - POST /api/users (Create a new user profile)
 - GET /api/users/{userId} (Fetch a user profile)
 - POST /api/resume/analyze (The main endpoint for the resume scorer)
 - POST /api/generate/answer (The endpoint for the tailored answer agent)
4. **Integrate Logging:** Add logging to your API endpoints from day one. Log incoming requests, important processing steps, and especially any errors. This will be invaluable for debugging.

Remember to start simple, and build up complexity iteratively. Good luck!

Note: There are a few helpful resource videos to reference in the main [Job_Assistant_Overview](#) page.

Data Scientist Assignment

Data Scientist Assignment

As a Data Scientist, you will design and build the "brain" of the AI Job Assistant. You will be responsible for the core intelligence of our product, creating the AI agents that scrape data, analyze documents, and generate valuable, human-like text to help users in their job search.

Phase 1: Preparation

The goal of this phase is to design the agentic workflows. You need to determine the tools, data, and LLM prompting strategies required to deliver on the product's core features.

Actions:

1. Research Core Technologies:

- **Web Scraping:** Read [Building a Job Scraper](#) to understand the challenges. Investigate Python libraries like **Beautiful Soup** and **Scrapy** for extracting content from job posting URLs.
- **Agent Frameworks:** Research an agentic framework like **LangChain** or **LlamaIndex**. Understand their core concepts of "Tools," "Chains," and "Agents." This will provide a structure for your work.
- **Agentic Design Patterns:** Focus your reading on patterns that are immediately relevant to this project: [Tool Use](#) and [Planning](#).

2. Design the Agentic Workflows: For each core feature, create a flowchart or diagram that details the step-by-step process.

- **Resume Scorer Workflow:**
 - **Input:** User's resume text, Job Description URL.
 - **Step 1 (Tool):** Call a `scrape_job_description` tool with the URL.
 - **Step 2 (LLM):** Feed the resume text and scraped job description text into a carefully designed prompt.
 - **Step 3 (Output Parsing):** Instruct the LLM to return a structured JSON object containing a score (integer) and suggestions (array of strings). Parse this output to ensure it's valid.
 - **Output:** The validated JSON object.
- **Tailored Answer Workflow:**
 - **Input:** User profile data, scraped job description text, the application question (e.g., "Why are you a good fit?").
 - **Step 1 (LLM):** Feed all three pieces of information into a prompt that instructs the LLM to act as a career coach and write a compelling answer.
 - **Output:** The generated text string.

3. **Define the Data Contracts:** Specify the exact structure of the data you'll be working with. This is crucial for successful integration with the backend team.
 - **Input Data:** What fields from the User Profile do you need? (e.g., work_history, skills, education).
 - **Output Data:** Define the precise JSON schema for the Resume Score output.

Office Hours (optional):

Use Office Hours to help with roadblocks rather than as a gatekeeper holding you back. This way you can start developing your code to make progress and no longer wait for Office hours.

Some things to think about:

- Your agent workflow diagrams.
- Your proposed data models/contracts.

Phase 2: Development

Can start development even if not attend Office hours. Start focusing on building the agent's core components. Your focus is on writing the functions for scraping, prompting, and parsing that the backend will eventually call.

Actions:

1. **Develop the Scraper Tool:** Write a robust function that takes a URL and returns the clean text of a job description. This is harder than it sounds. Account for things like navigation bars, ads, and footers. Test it on several different job sites (e.g., LinkedIn, Indeed, Greenhouse).
2. **Prototype the LLM Prompts:** Write the prompt templates for your agents. This is an iterative process. Start with a basic prompt, and refine it to get better, more consistent results. Use a simple script to test your prompts.
 - **Key Tip:** Use "few-shot" prompting. In your prompt, provide an example of the input and the ideal output format you want. This dramatically improves the reliability of structured data output (like JSON).
3. **Build the Agent Service Function:** Create a primary function, like `analyze_resume_and_jd`, that encapsulates the entire workflow: it should call your scraper tool, format the prompt with the data, call the LLM, and parse the response.
4. **Set Up for Integration:** Ensure your Python code is organized into functions that can be easily imported and called by the backend server (e.g., the Flask/Express app).



Remember to start simple and iterate. Focus on getting a basic end-to-end flow working before optimizing or adding complex features.

Note: There are a few helpful resource videos to reference in the main [Job_Assistant_Overview](#) page.